Algebra 2 Functions & Transformations Cheat Sheet

Unit Focus: *function notation, parent functions, and transformations*

1 KEY VOCABULARY

function	A relation (set of ordered pairs) where each input x has **exactly one** output y.
domain	The set of all allowed inputs (x-values).
range	The set of all outputs (y-values).
function notation	$f(x)$ is read "f of x": the output when the input is x. It does **not** mean f times x.
parent function	The simplest function of a family, such as $y = x^2$. Transformations start from it.
piecewise function	A function with different rules on different parts of the domain.
composite function	$(f \circ g)(x) = f(g(x))$. The output of g becomes the input of f.
inverse function	f^{-1} undoes f: inputs and outputs swap, and the graph reflects over $y = x$.
one-to-one	Every output comes from exactly one input (passes the horizontal line test). Only these have inverses.

2 FUNCTION NOTATION, DOMAIN & RANGE

Example: $f(x) = 2x^2 - 3x + 1$

$f(-2) = 2(-2)^2 - 3(-2) + 1 = 8 + 6 + 1 = \mathbf{15}$ (use parentheses!)

$f(a + 1) = 2(a + 1)^2 - 3(a + 1) + 1 = \mathbf{2a^2 + a}$

$x \geq 2 \rightarrow [2, \infty)$	$x < 5 \rightarrow (-\infty, 5)$
$-1 < x \leq 4 \rightarrow (-1, 4]$	$x \neq 3 \rightarrow (-\infty, 3) \cup (3, \infty)$

[ ] includes the endpoint, ( ) excludes it; ∞ always gets ( ). $f(3)$ is the y-value at $x = 3$ on a graph or table.

Square root: radicand ≥ 0
$g(x) = \sqrt{x - 2} + 1$
$x - 2 \geq 0$: D: $[2, \infty)$, R: $[1, \infty)$

Fraction: denominator ≠ 0
$h(x) = \dfrac{3}{x + 4}$
$x \neq -4$: D: $(-\infty, -4) \cup (-4, \infty)$

3 RELATIONS & THE VERTICAL LINE TEST

Function test: no x may have two different y-values. Different xs **may** share a y.

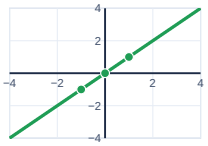
Function: {(1, 3), (2, 5), (4, 5)}
$y = x^2$ (each x gives one y)

Not a function: {(1, 3), (1, 7), (2, 5)}
$x = y^2$: $x = 4$ gives $y = 2$ and -2

Vertical line test: if any vertical line crosses a graph **more than once**, it is not a function.

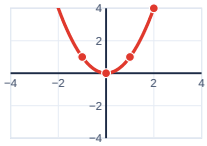
4 PARENT FUNCTION GALLERY

Linear



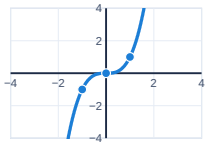
$y = x$
D: $(-\infty, \infty)$
R: $(-\infty, \infty)$

Quadratic



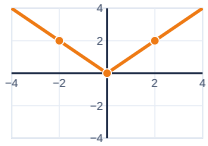
$y = x^2$
D: $(-\infty, \infty)$
R: $[0, \infty)$

Cubic



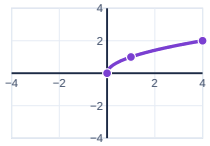
$y = x^3$
D: $(-\infty, \infty)$
R: $(-\infty, \infty)$

Absolute value



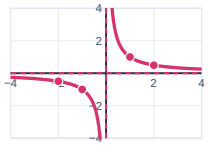
$y = |x|$
D: $(-\infty, \infty)$
R: $[0, \infty)$

Square root



$y = \sqrt{x}$
D: $[0, \infty)$
R: $[0, \infty)$

Reciprocal



$y = \dfrac{1}{x}$
D: $x \neq 0$
R: $y \neq 0$

Dots mark points on each graph. D = domain, R = range. The reciprocal graph has a vertical asymptote at $x = 0$ and a horizontal asymptote at $y = 0$ (dashed).

5 TRANSFORMATION RULES

$$g(x) = a \cdot f(b(x - h)) + k$$

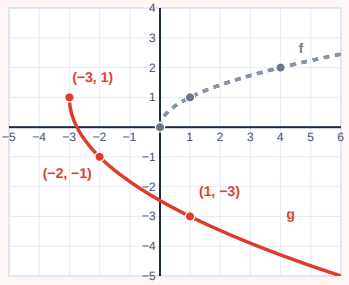
$f(x) + k$	up k (down if $k < 0$); changes y
$f(x - h)$	right h. **$f(x + 3)$ moves LEFT 3**
$-f(x)$	reflect in the x-axis: $(x, y) \rightarrow (x, -y)$
$f(-x)$	reflect in the y-axis: $(x, y) \rightarrow (-x, y)$
$a \cdot f(x)$	$\lvert a \rvert > 1$ vertical stretch; $0 < \lvert a \rvert < 1$ vertical compression
$f(bx)$	$\lvert b \rvert > 1$ horizontal compression (by $\frac{1}{b}$); $0 < \lvert b \rvert < 1$ horizontal stretch

Order: work inside to outside: b (stretch or reflect), then h (shift), then a (stretch or reflect), then k (shift).

6 TRANSFORMATION EXAMPLE

Example: Graph $g(x) = -2\sqrt{x + 3} + 1$ from $f(x) = \sqrt{x}$

1. $x + 3$: left 3
2. -2: reflect in x-axis, stretch by 2
3. $+ 1$: up 1
Rule: $(x, y) \rightarrow (x - 3, -2y + 1)$
$(0, 0) \rightarrow (-3, 1)$
$(1, 1) \rightarrow (-2, -1)$
$(4, 2) \rightarrow (1, -3)$
D: $[-3, \infty)$ **R:** $(-\infty, 1]$



7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:
- Horizontal changes act on x and do the **opposite**: $f(x + 3)$ shifts left.
- Multiply by a before adding k.
- Track key points with $(x, y) \rightarrow$ (new x, new y).
- A square root needs a radicand ≥ 0; a fraction needs a denominator ≠ 0.

COMMON MISTAKES:
- Shifting $f(x + 3)$ to the right.
- Mixing up $f(x) + 3$ (up 3) and $f(x + 3)$ (left 3).
- Mixing up $-f(x)$ (over the x-axis) and $f(-x)$ (over the y-axis).
- Reading $f(x)$ as f times x.

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More Practice: *operations, composition, piecewise, and inverse functions*

1 OPERATIONS ON FUNCTIONS

Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$.

Sum	$(f + g)(x) = f(x) + g(x) = \mathbf{x^2 + 2x - 2}$
Difference	$(f - g)(x) = (2x + 1) - (x^2 - 3) = \mathbf{-x^2 + 2x + 4}$
Product	$(f \cdot g)(x) = (2x + 1)(x^2 - 3) = \mathbf{2x^3 + x^2 - 6x - 3}$
Quotient	$(\frac{f}{g})(x) = \frac{2x+1}{x^2-3},\ x \neq \pm\sqrt{3}$

Domain: x-values allowed in **both** functions; for a quotient, also $g(x) \neq 0$.
Evaluate: $(f + g)(2) = f(2) + g(2) = 5 + 1 = \mathbf{6}$
Watch out: $(f \cdot g)(x)$ multiplies the outputs; $(f \circ g)(x)$ puts g inside f. They are different.

2 COMPOSITION OF FUNCTIONS

$$(f \circ g)(x) = f(g(x))$$

Work from the **inside out**: do g first, then feed its output into f.

$f(g(x)) = 2(x^2 - 3) + 1$
$(f \circ g)(x) = \mathbf{2x^2 - 5}$
$(f \circ g)(2) = f(g(2)) = f(1) = \mathbf{3}$

$g(f(x)) = (2x + 1)^2 - 3$
$(g \circ f)(x) = \mathbf{4x^2 + 4x - 2}$
Order matters: $f \circ g \neq g \circ f$.

x	1	2	3	4
$g(x)$	3	1	4	
$f(x)$	5	0	2	7

$(f \circ g)(2) = f(g(2)) = f(1) = \mathbf{5}$ $\quad (g \circ f)(3) = g(f(3)) = g(2) = \mathbf{1}$

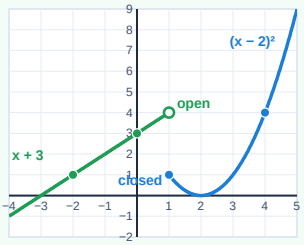
Domain: x must be in the domain of g, and $g(x)$ in the domain of f. For $f = \sqrt{x}$, $g = x - 4$: $(f \circ g)(x) = \sqrt{x - 4}$, $x \geq 4$.

3 PIECEWISE FUNCTIONS

$$\begin{cases} x + 3, & \text{if } x < 1 \\ (x - 2)^2, & \text{if } x \geq 1 \end{cases}$$

Evaluate: pick the rule whose condition the input satisfies.
$h(0)$: $0 < 1$, so $0 + 3 = \mathbf{3}$
$h(1)$: $1 \geq 1$, so $(1 - 2)^2 = \mathbf{1}$
$h(4)$: $(4 - 2)^2 = \mathbf{4}$

Graph: graph each piece only on its interval. **Open circle** for < or >; **closed circle** for ≤ or ≥. At $x = 1$ the left piece ends at an open circle (1, 4); the right piece starts at a closed circle (1, 1).



4 INVERSES, GRAPHS & ONE-TO-ONE

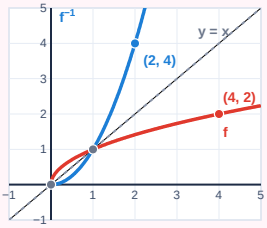
(a, b) on $f \Leftrightarrow (b, a)$ on f^{-1}. The graphs are mirror images over $y = x$. Domain and range **swap**.

Example: $f(x) = \sqrt{x}$, $f^{-1}(x) = x^2$, $x \geq 0$

$(4, 2)$ on $f \rightarrow (2, 4)$ on f^{-1}

Horizontal line test: if any horizontal line hits the graph twice, f is not one-to-one and has no inverse. $y = x^2$ fails (2 and −2 give 4) until you restrict to $x \geq 0$.

$$f^{-1}(x) \neq \frac{1}{f(x)}$$



5 FINDING & VERIFYING AN INVERSE

Example: Find the inverse of $f(x) = 3x - 5$.

1. Write $y = 3x - 5$.
2. Swap x and y: $x = 3y - 5$.
3. Solve for y: $3y = x + 5$, $y = \frac{x+5}{3}$.
4. Name it: $f^{-1}(x) = \frac{x+5}{3}$

Verify with composition (both must equal x):
$f(f^{-1}(x)) = 3 \cdot \frac{x+5}{3} - 5 = (x + 5) - 5 = x$ ✓
$f^{-1}(f(x)) = \frac{(3x - 5) + 5}{3} = \frac{3x}{3} = x$ ✓

From a table: if $f(3) = 7$, then $f^{-1}(7) = 3$. **Symmetry (optional):** even if $f(-x) = f(x)$ (y-axis, like x^2); odd if $f(-x) = -f(x)$ (origin, like x^3).

6 WRITING THE EQUATION OF A GRAPH

1. Name the parent shape. **2.** Read the key point (vertex, corner, or start) for h and k. **3.** Use one more point to find a. **4.** Check a point.

Example: A parabola has vertex (1, 4) and passes through (0, 2).

$y = a(x - 1)^2 + 4$ (vertex form, $h = 1$, $k = 4$)
Use (0, 2): $2 = a(0 - 1)^2 + 4 \rightarrow 2 = a + 4 \rightarrow a = -2$
$y = -2(x - 1)^2 + 4$ Check $x = 2$: $-2(2 - 1)^2 + 4 = 2$, so (2, 2) is on it ✓

Example: Absolute value with vertex (−2, −1) through (0, 3)

$y = a|x + 2| - 1$; $3 = a|2| - 1 \rightarrow a = 2$. $y = 2|x + 2| - 1$

Clues: opens down → $a < 0$. Shift left → h is negative inside: $x + 2$ means $h = -2$. Same shape as $y = x^2$ but narrower → $|a| > 1$.

7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:
- $(f \circ g)(x)$ means g goes **first** (inside).
- To find an inverse: swap x and y, then solve for y.
- Check inverses by composing both ways.
- Open circle for < and >; closed for ≤ and ≥.
- Check that each x falls in exactly one piece.
- Restrict a domain (like $x \geq 0$) to make $y = x^2$ one-to-one.

COMMON MISTAKES:
- Reading $f \circ g$ as "f first".
- Writing $f^{-1}(x) = \frac{1}{f(x)}$. The −1 is not an exponent.
- Forgetting to swap x and y.
- Assuming $f(a + b) = f(a) + f(b)$.
- Finding a domain without checking roots and denominators.
- Using a closed circle for < or > at a piece boundary.